



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 7

1. Case Study Title: To demonstrate events handling in javascript, a) access and manipulate from elements using javascript. b) validate user input (eg. email,empty,fields) c) use various form events (submit,change,focus).

2. Case Study Program Code:

```
//Sonal Nakhate 24070521199
```

```
<!DOCTYPE html>
<html>
<head>
  <title>Sonal's - Event Handling</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #e8f0fe;
      padding: 40px;
    }

    .form-box {
      width: 450px;
      margin: auto;
      background-color: white;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 4px 12px gray;
    }
  </style>
</head>
<body>
```



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```
h2 {
    text-align: center;
}

label {
    display: block;
    margin-top: 15px;
    font-weight: bold;
}

input, select {
    width: 100%;
    padding: 10px;
    margin-top: 5px;
    box-sizing: border-box;
}

input:focus, select:focus {
    border: 2px solid blue;
    outline: none;
}

.error {
    color: red;
    font-size: 13px;
}

button {
    width: 100%;
    margin-top: 20px;
    padding: 10px;
    background-color: #2563eb;
    color: white;
```



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```
        border: none;
        cursor: pointer;
    }

    #message {
        text-align: center;
        font-weight: bold;
        margin-top: 15px;
    }
</style>
</head>

<body>

<div class="form-box">

    <h2>Sonal's student Registration Form</h2>

    <form id="studentForm">

        <label>Name:</label>
        <input type="text" id="name">
        <div id="nameError" class="error"></div>

        <label>Email:</label>
        <input type="text" id="email">
        <div id="emailError" class="error"></div>

        <label>Course:</label>
        <select id="course">
            <option value="">Select Course</option>
            <option value="CSE">Computer Science</option>
            <option value="IT">Information Technology</option>
            <option value="ECE">Electronics</option>
```



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```
</select>

<div id="courseError" class="error"></div>

<button type="submit">Submit</button>

</form>

<p id="message"></p>

</div>

<script>

    // Access form elements
    let form = document.getElementById("studentForm");
    let name = document.getElementById("name");
    let email = document.getElementById("email");
    let course = document.getElementById("course");

    // Focus event
    name.addEventListener("focus", function() {
        name.style.backgroundColor = "#eef4ff";
    });

    email.addEventListener("focus", function() {
        email.style.backgroundColor = "#eef4ff";
    });

    // Change event
    course.addEventListener("change", function() {
        document.getElementById("courseError").innerText =
            "Selected Course: " + course.value;
    });
```



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```
// Submit event
form.addEventListener("submit", function(event) {

    event.preventDefault();

    let valid = true;

    // Name validation
    if (name.value.trim() === "") {
        document.getElementById("nameError").innerText =
            "Name cannot be empty";
        valid = false;
    } else {
        document.getElementById("nameError").innerText = "";
    }

    // Email validation
    let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

    if (email.value.trim() === "") {
        document.getElementById("emailError").innerText =
            "Email cannot be empty";
        valid = false;
    }
    else if (!emailPattern.test(email.value)) {
        document.getElementById("emailError").innerText =
            "Enter a valid email";
        valid = false;
    }
    else {
        document.getElementById("emailError").innerText = "";
    }

    // Course validation
```



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```
if (course.value === "") {
    document.getElementById("courseError").innerText =
        "Please select a course";
    valid = false;
}

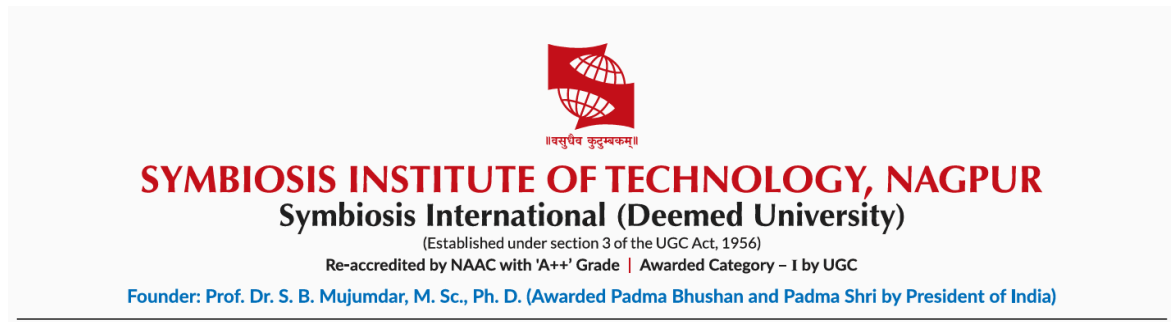
// Final result
if (valid) {
    document.getElementById("message").innerText =
        "Registration submitted successfully!";
    document.getElementById("message").style.color = "green";
}
else {
    document.getElementById("message").innerText =
        "Please correct the errors.";
    document.getElementById("message").style.color = "red";
}

});

</script>

</body>
</html>
```

3. Output:

A screenshot of a web browser window. The address bar shows the file path 'C:/Users/Sonal/Desktop/Documents/JS_24070521199/Exp7/CS.html/cs.html'. The page content is a white form titled 'Sonal's student Registration Form' centered on a light blue background. The form has three input fields: 'Name:' with 'Sonal Nakhate', 'Email:' with 'abc@xyz.com', and 'Course:' with a dropdown menu showing 'Computer Science'. Below the dropdown, it says 'Selected Course: CSE' in red. A blue 'Submit' button is at the bottom of the form, and below it, the text 'Registration submitted successfully!' is displayed in green.

4. Result/Conclusion: Thus, JavaScript events were successfully handled using focus, change, and submit, and form elements were accessed, manipulated, and validated using JavaScript.